

SESSION 8 — LOOPS

Why Do We Need Loops?

Imagine: You want to print numbers from 1 to 10.

Without loop: You would write `print()` 10 times.

Loop allows:

- Repeating code
- Automating repetition
- Working with collections (lists)

There are two main types of loops in Python:

1. for loop
2. while loop

We will learn both step by step.

✓ FOR LOOP

A for loop is used when: You know how many times you want to repeat something.

Basic Syntax:

for variable in sequence: statement

- variable changes automatically
- sequence can be list, range, string, etc.

```
# Print numbers from 1 to 5

for i in range(1,6):
# for i in range(3):
    print(i)
```

```
1
2
3
4
5
```

```
# Print numbers from 1 to 5

for i in range(0,6):
    print(i)
    print(i+1)
    print(i+2)
    print(i+3)
    print(i+4)
    print("-----")
```

```
0
1
2
3
4
-----
1
2
3
4
5
-----
2
3
4
5
6
-----
```

```

3
4
5
6
7
-----
4
5
6
7
8
-----
5
6
7
8
9
-----

```

```
sales = [1000, 2000, "1ddd500", 3000]
```

```
for amount in sales:
    print("Sale amount:", amount)
```

```

Sale amount: 1000
Sale amount: 2000
Sale amount: 1ddd500
Sale amount: 3000

```

✓ ENUMERATE

enumerate() gives:

- index
- value

Used when you need position and value both.

```
sales = [1000, 2000, 1500]
```

```
for index, amount in enumerate(sales):
    print("Day", index, "Sale:", amount)
```

```

Day 0 Sale: 1000
Day 1 Sale: 2000
Day 2 Sale: 1500

```

```
sales = [1000, 2000, 1500, 3000]
```

```
total = 0
```

```

for amount in sales:
    total = total + amount
    # total += amount

    print(total)

```

```
print("Total Sales:", total)
```

```

1000
3000
4500
7500
Total Sales: 7500

```

1. Write a `for` loop that prints numbers from 1 to 5.
2. Write a `for` loop that prints all even numbers from 1 to 20.
3. Write a `for` loop that prints each element in the list: [10, 20, 30, 40, 50].
4. Write a `for` loop that finds the sum of numbers from 1 to 10.
5. Write a `for` loop that prints the multiplication table of 7.
6. Write a `for` loop that counts how many numbers in the list [2, 5, 8, 11, 14] are greater than 6.